

Dear Gatik team,

I'm applying for the Staff Engineer, Perception role because the combination you're looking for - classical computer vision, modern ML, 3D geometry, optimized C++, and taking perception from R&D through production - closely matches the work I've spent much of my career doing.

I co-founded Cambrian and led its technical work across computer vision, ML, GPU computing, and real-time systems. We built a perception and mixed-reality platform that shipped to millions of users and later contributed to automotive computer-vision programs.

Before ARKit or similar perception platforms existed, we had to build the underlying machinery ourselves: feature and line detection, image filtering, segmentation, tracking, optical flow, camera pose and calibration, RANSAC plane fitting, multi-frame fusion, depth and surface-normal estimation, and experiments in SLAM and 3D scene reconstruction. I also worked on automotive mixed-reality HUD problems involving fast camera-pose estimation, geometry, and sensor fusion in C++, including execution under tight CPU constraints.

Our learned perception components presented another systems problem. Segmentation and surface-normal CNNs could take 1-3 seconds on the mobile hardware of the time, while the scene still had to evolve at frame rate. We built a priority-scheduled, backpressured C++ runtime that combined asynchronous learned results with tracking, geometry, temporal evidence, and sensor input without allowing the slowest estimator to stall the world model.

That work crossed the full R&D-to-production path: novel segmentation methods, CNN training, synthetic data generation in Unreal Engine, model quantization and compression, TensorFlow and other inference targets, OpenCV, SIMD, GPU/GPGPU acceleration, and production SDKs across multiple platforms. I am also first-named inventor on a patent covering our segmentation work.

My recent work has expanded into transformer architecture, CUDA/Metal inference, model pruning and quantization, distributed AI, and constrained inference runtimes. The scale and models have changed, but the engineering problem I enjoy is much the same: understand the complete system, find where the assumptions break, and make difficult computation work efficiently on real hardware.

I haven't spent my career inside the autonomous-vehicle industry, and my perception background is more camera-centric than Gatik's image/LIDAR stack. What I bring is deep experience across the classical vision, learned perception, 3D geometry, real-time systems, and performance disciplines underneath it, along with a long track record of turning research into deployed products.

I'd be excited to bring that combination to Gatik's perception team.

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